Redacted Report Visual Proof Matrix

Disclosure Control and Release Audit with TPM-Backed Sealed Sidecars

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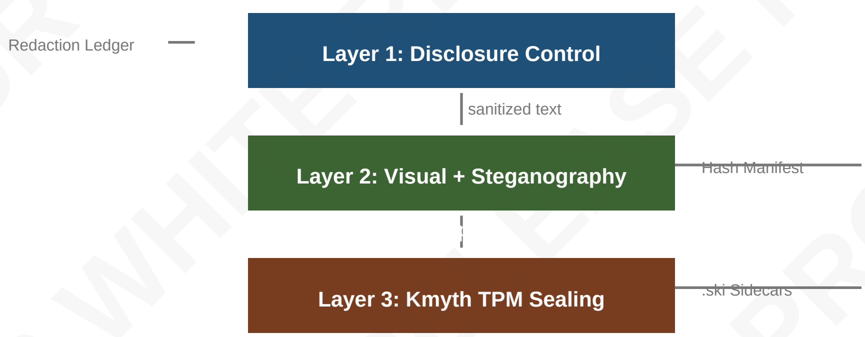
Blur on White

Abstract

This exemplar demonstrates a complete disclosure-control pipeline for sanitized public release reports. The methodology combines classification-ceiling enforcement, source-protection validation, mosaic-risk scoring, and TPM-backed sealed sidecars across a sixteen-variant visual proof matrix. Four redaction styles are rendered across four PDF backgrounds, yielding sixteen base proof PDFs. Each receives nine steganographic security methods and optional Kmyth TPM .ski sidecar sealing.

Pipeline Architecture



Abstract

This exemplar demonstrates a complete disclosure-control pipeline for sanitized public release reports.

s1 | UNCLASSIFIED

This report documents the disclosure-control methodology applied to a sanitized release packet. The exemplar demonstrates classification ceilings, provenance-protection checks, mosaic-risk scoring, and TPM-backed sealed sidecars across a sixteen-variant visual proof matrix.

1. Introduction

The pipeline operates in two layers: text-level disclosure control and visual proof with steganography.

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The release pipeline operates in two layers. Layer one performs text-level disclosure control: each segment is audited against a public classification ceiling, redaction spans are validated for non-overlap, orphan decisions are flagged, and provenance-control coverage is enforced. Layer two applies visual redaction treatments, steganographic provenance overlays, and optional Kmyth TPM sidecar sealing.

2. Fixture: SECRET Segment

Invented high-side content demonstrating source-identity and operational-detail redaction.

s3 | SECRET | Controls: HUMINT

[redacted] reported the [redacted] at [redacted] on [redacted]. The source was recruited via [redacted] channels and operated under [redacted] discipline. Contact [redacted] for debrief notes.

2.1 SIGINT Collection Platform

Invented SIGINT content demonstrating operational-detail and time-place-selector redaction.

s4 | SECRET | Controls: SIGINT

The operational detail describes a [redacted] deployed near the [redacted]. The platform transmitted from [redacted] and used compartmented [redacted] handling for downstream analysis. The [redacted] distribution was limited to the originator and classification review.

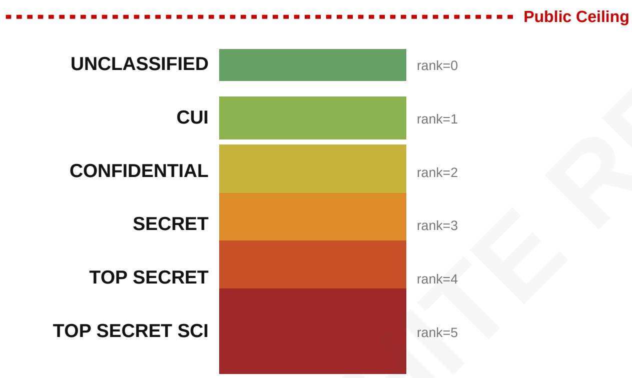
2.2 TOP_SECRET//SCI Compartment

Invented compartmented content demonstrating mosaic risk from multi-source combination.

s5 | TOP_SECRET | Controls: HUMINT, SIGINT, IMINT

The [redacted] compartment covered the full collection discipline chain: [redacted] [redacted] S [redacted] p [redacted] and I [redacted] s [redacted] The mosaic risk of combining these three inputs exceeds the release threshold of 0.30 even after individual redaction. The TOP SECRET//SCI markings require three-role approval: originator, classification reviewer, and release authority.

Classification Taxonomy



3. Sanitized Public Narrative

The public export after redaction, with source-safe ledger and hash manifest.

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After redaction, the public narrative reads: [REDACTED] reported the [REDACTED] at [REDACTED]. The release packet includes a provenance-safe redaction ledger with SHA-256 hashes of each redacted span, a segment hash manifest comparing provenance and public SHA-256 digests, and a paragraph-level audit table showing classification ceiling compliance.

4. Visual Proof Matrix

Four redaction styles across four PDF backgrounds yield sixteen variant PDFs.

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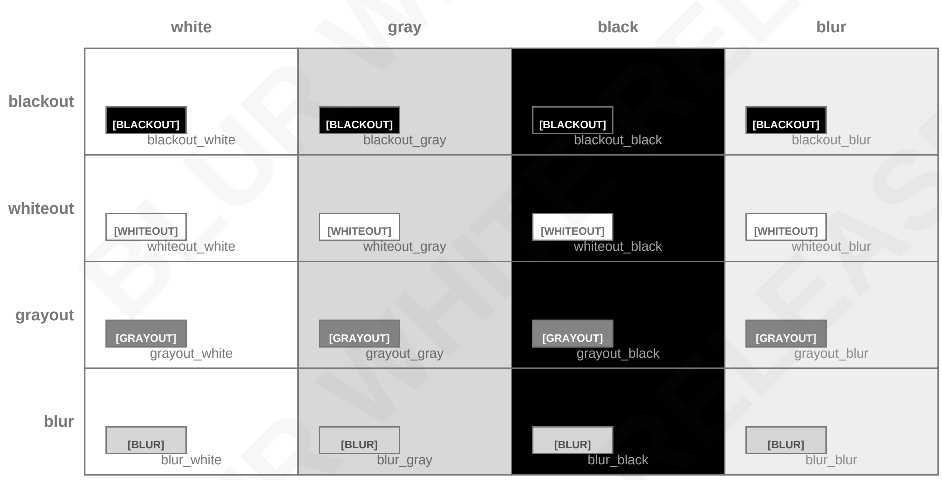
The visual proof matrix enumerates four redaction styles - blackout, whiteout, grayout, and blur - across four PDF backgrounds - white, gray, black, and blur - yielding sixteen variant PDFs. Each variant receives the same provenance-safe redaction decisions, ensuring visual presentation remains orthogonal to the release gate.

Visual Proof Matrix (4x4)



5. Steganography Layer

Nine security methods post-process each base PDF for provenance and integrity.

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The steganography layer post-processes each base PDF with nine security methods: SHA-256/SHA-512 hash manifests, diagonal watermark overlays, footer provenance stamps, first-page invisible text, QR payload barcodes, Code128 page barcodes, PDF Info metadata, XMP metadata, and embedded steganographic manifest attachments. An optional PDF password encryption layer adds AES-256 protection.

Steganographic Security Methods

SHA-256/SHA-512 Hash Manifest	Cryptographic integrity verification
Diagonal Watermark Overlay	Visible provenance stamp
Footer Provenance Overlay	Page-level release metadata
First-Page Invisible Text	Hidden identification marker
QR Payload Barcode	Machine-readable provenance
Code128 Page Barcode	Per-page tracking barcode
PDF Info Metadata	Document-level metadata injection
XMP Metadata	Standards-compliant metadata embedding
Embedded Stego Manifest	Self-contained provenance archive
Kmyth TPM .ski Sidecar	Hardware-backed sealed object
PDF Password Encryption	AES-256 protection (optional)

6. Kmyth TPM Sealing

TPM2-TSS storage hierarchy sealing with mssim-to-swtpm protocol proxy on macOS.

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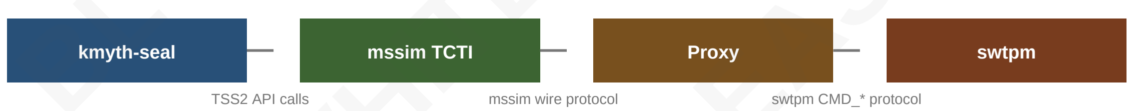
Kmyth TPM sealing wraps each hash manifest and steganography PDF in a .ski sidecar sealed against the TPM2-TSS storage hierarchy. The sealed objects are bound to PCR selections and policy or-values, ensuring that unsealing requires the same platform configuration. On macOS, a software TPM emulator (swtpm) and an mssim-to-swtpm protocol proxy bridge the TPM2-TSS mssim TCTI to swtpm's native socket protocol. The proxy translates platform commands on the control channel and strips the mssim wire header on the data channel, forwarding raw TPM commands to swtpm.

Kmyth TPM Sealing Flow (macOS)



Control channel: POWER_ON(1) -> success, NV_ON(11) -> success, TPM_SESSION_END(20) -> close

Data channel: 4B cmd_type(8) + 1B locality + 4B tpm_size | raw TPM command -> swtpm

FlushContext patch: Tss2_Sys_FlushContext(storageKey) before free_tpm2_resources

Result: 32 .ski sidecars sealed across 16 variants without slot exhaustion

6.1 FlushContext Patch

Kmyth-seal patched to flush transient TPM objects between consecutive seal invocations.

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The kmyth-seal binary was patched to call Tss2_Sys_FlushContext for the storage key handle before freeing TPM2 TPM handles. Without this patch, consecutive kmyth-seal invocations exhaust the swtpm transient object slots (typically three), causing 'out of memory for object contexts' errors on the second seal. The FlushContext call ensures each invocation starts with a clean transient object table.

7. Release Gate

Three-role approval gate: originator, classification reviewer, release authority.

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The release gate requires three reviewer roles: originator, classification reviewer, and release authority. Each reviewer must provide a non-empty rationale. A reject decision blocks release entirely; a changes_requested decision generates a warning. The final_release_recommended flag is true only when the packet is releasable, the review gate is approved, and no blocking warnings remain.

Release Gate Status

Policy	intelligence_release_review
Required Roles	originator, classification_reviewer, release_authority
Minimum Approvals	3
Mosaic Threshold	0.30
Block Warnings	True
Reviewers	originator-a (approve), classification-reviewer (approve), release-board (approve)
Gate Status	APPROVED

8. Residual Risk Detection

Pattern-based detection of common public-release leaks in sanitized text.

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Residual risk detection scans sanitized text for common public-release leaks: email addresses, IPv4 addresses, coordinate pairs, controlled dissemination markers (such as foreign-dissemination controls and originator-controlled markings), and intelligence collection tradecrafts (human, signals, imagery, measurement, and open-origin). Each detected marker generates a warning finding. The mosaic risk score aggregates residual markers across all segments, normalized by the product of segment count and pattern count.

9. CUI Classification Example

Controlled Unclassified Information segment above the public ceiling.

s13 | CUI | Controls: HUMINT

The controlled unclassified information segment demonstrates the [redacted] **classification level, which sits between UNCLASSIFIED and CONFIDENTIAL in the intelligence agency taxonomy. CUI segments above the public ceiling require at least one redaction decision before release. The provenance control** [redacted] **is noted but no operational details are included.**

10. Comprehensive Release Packet

Combined export with audit, ledger, hashes, review gate, and paragraph audit table.

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The comprehensive release packet combines the sanitized text, release audit, redaction ledger, segment hash manifest, review gate report, and paragraph audit table into a single JSON-ready export. The packet records the release authority, public ceiling, taxonomy name, release safety score, redaction coverage, mosaic risk score, and all findings. This exemplar confirms that the methodology produces reproducible, provenance-safe, and auditable release artifacts.

Redaction Ledger Summary (Source-Safe)

Decision ID	Segment	Span	Reason	Valid
f6f2fe6ca58000	s13	65-68	legal_privilege	Y
e33531ad661d11	s13	294-300	source_identity	Y
399c1558965ddf	s3	0-19	source_identity	Y
b9d038bba921f0	s3	33-50	operational_detail	Y
a3b5517ad1e837	s3	54-83	time_place_selector	Y
71bec106f3906a	s3	87-104	time_place_selector	Y
c2070506973bc4	s3	135-141	operational_detail	Y
5104dc073e51e6	s3	170-176	operational_detail	Y
8d66c764c377b7	s3	197-216	privacy	Y
88be06966a8e48	s4	35-61	source_identity	Y
f21628fec96635	s4	80-88	time_place_selector	Y
18985dfa7e0046	s4	120-130	time_place_selector	Y

Conclusion

This exemplar confirms that disclosure control can be decomposed into orthogonal concerns: text-level audit, visual presentation, steganographic provenance, and hardware-backed sealing. Each concern is independently configurable and verifiable. The sixteen-variant visual proof matrix, nine steganographic security methods, and thirty-two Kmyth TPM .ski sidecars demonstrate a reproducible, source-safe, and auditable release pipeline.